

MATH 231 – Applied Linear Algebra

Lecture 1 ■ Vectors: Magnitude, Direction, and the First Vector Operations

Tuesday, September 1, 2026 ■ Kiely Hall 326 ■ Instructor: Kevin Bedoya

What you should be able to do afterwards. State what a vector is in terms of magnitude and direction; identify the tail and head of a vector in the plane; compute the Euclidean norm of a vector and explain why it is a *unary* operation; distinguish vector quantities from scalar quantities; read all three notational conventions for a vector; add and subtract vectors in the plane both componentwise and geometrically; and scale a vector by a real number, saying what the scaling does to its magnitude and direction.

Further reading. Boyd & Vandenberghe, *Introduction to Applied Linear Algebra*, Ch. 1 (vectors, addition, scalar multiplication) and Ch. 3 (norm and distance); Strang, *Linear Algebra and Its Applications*, Ch. 1; Axler, *Linear Algebra Done Right*, Ch. 1; Trefethen & Bau, *Numerical Linear Algebra*, Lec. 2–3 (norms).

1. Vectors: magnitude, direction, and the space they live in

We begin with a working definition — deliberately informal, and stated in the language of quantities rather than axioms.

Definition 1 (Vector, working definition). A *vector* is a mathematical object carrying two attributes: a **magnitude** and a **direction**.

Looking ahead. This is the physicist’s definition, and it is the right one to start from because it is the one you can *see*. Later in the course we replace it with the algebraist’s definition: a vector will become any element of a set equipped with two operations obeying a short list of axioms — at which point functions, polynomials, and matrices all become vectors too, and “magnitude and direction” stops being the defining feature. For now, hold on to the picture.

The space a vector lives in

A vector does not exist in isolation; it lives in a space. Ours today is the **two-dimensional Euclidean plane**, the familiar xy -plane, and every vector we draw will live in it.

The word *dimension* here means a **degree of freedom** with respect to spatial coordinates: how many independent numbers must you supply to pin down a location? On a line, one; on the plane, two — an x and a y , chosen independently of one another. The plane is two-dimensional in exactly that sense.

Tail, head, and the two attributes

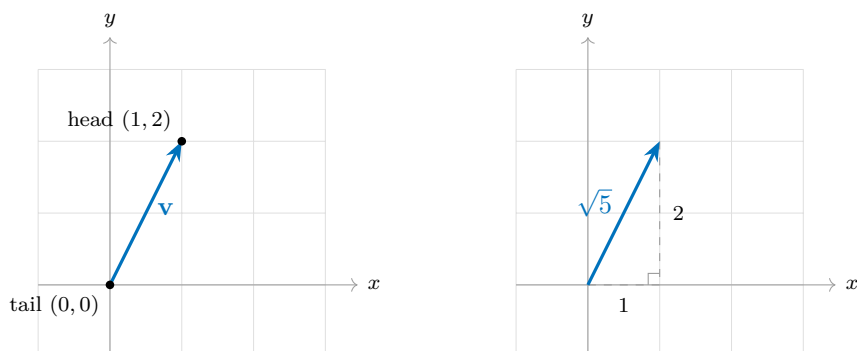
Fix the point $(1, 2)$. Draw an arrow from the origin to it. That arrow is a vector, and it has two distinguished points:

- the **tail** — where the arrow begins, here the origin $(0, 0)$;
- the **head** — where the arrow ends and the arrowhead is drawn, here $(1, 2)$.

With the tail and head named, the two attributes become concrete:

- **Direction** is the arrow's orientation in the plane — which way it points, relative to the axes, as it travels from tail to head. The arrow to $(1, 2)$ points up and to the right, more steeply up than rightward.
- **Magnitude** is the arrow's **length**: the distance from the tail to the head. We measure it with the Euclidean distance, and we compute it in §2.

Note the arrowhead in the left-hand figure below: the arrowhead *is* the direction, so a bare line segment would lose half the information.



2. The norm — our first vector operation

The magnitude of a vector is not just a description of it; it is something we *compute* from it. Computing it is our first vector operation, and it has a name.

Definition 2 (Euclidean norm). For a vector $\mathbf{v}$ in the plane with tail at the origin and head at (x, y) , the **Euclidean norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{x^2 + y^2},$$

the Euclidean distance from the tail to the head.

Two things to notice about the shape of this operation.

- It is **unary**: it takes *one* vector as input. Compare $3 + 5$, where addition consumes two numbers — a *binary* operation. The norm consumes one vector and returns one number.
- It turns a *vector* into a *scalar*: the input is a vector in the plane, the output is a single real number, and that number is never negative. Length cannot be negative.

The formula is Pythagoras, nothing more. The right-hand figure above is the proof: the horizontal leg has length x , the vertical leg has length y , and the vector is the hypotenuse.

Example 1 (The running vector). For $\mathbf{v}$ with head at $(1, 2)$,

$$\|\mathbf{v}\| = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5} \approx 2.236.$$

Note that $\sqrt{5}$ is a finished answer: a norm is generically irrational, and there is no need to convert it to a decimal.

Example 2 (A clean one). For $\mathbf{w}$ with head at $(3, 4)$, $\|\mathbf{w}\| = \sqrt{9 + 16} = \sqrt{25} = 5$.

Remark. So three phrases now name the same number: the *magnitude* of the vector, the *length* of the arrow, and the *Euclidean norm* $\|\mathbf{v}\|$. Use them interchangeably.

3. Bridge to calculus: the derivative as a vector quantity

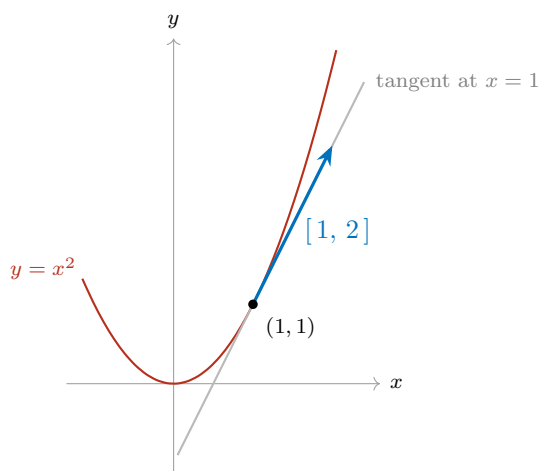
Vectors are not new to you — you have been computing one all through calculus without calling it that.

Let f be differentiable and consider the tangent line to the curve $y = f(x)$ at a point. The tangent line has both of our attributes:

- **Direction.** The *sign* of the slope orients it. A positive slope tilts up-and-to-the-right and the function is increasing there; a negative slope tilts down-and-to-the-right and the function is decreasing. The tangent line points somewhere.
- **Magnitude.** The *size* of the slope measures how fast the function is changing — the “speed” of change in that direction. A slope of 10 climbs ten times as urgently as a slope of 1.

Direction plus magnitude: the derivative of f at a point can be read as a **vector quantity**.

Concretely, take $f(x) = x^2$ at $x = 1$. Then $f'(1) = 2$, so from the point $(1, 1)$ the tangent advances 1 unit rightward for every 2 units upward: its direction vector is $[1, 2]$ — the very vector we drew in §1, now carrying a meaning. Its norm, $\sqrt{5}$, is the rate at which arc length accumulates along the tangent per unit of horizontal travel.



4. Vector quantities vs. scalar quantities

The phrase from §3 is worth making official, because it sorts most of the quantities you will meet in the sciences into two bins.

- A **vector quantity** has a magnitude *and* a direction. *Examples:* displacement, velocity, acceleration, force, momentum, torque, the electric and magnetic fields, a temperature *gradient*.
- A **scalar quantity** has a magnitude only — it indicates *scale*, and nothing else. *Examples:* mass, speed, temperature, energy, work, electric charge, time, pressure, density, volume.

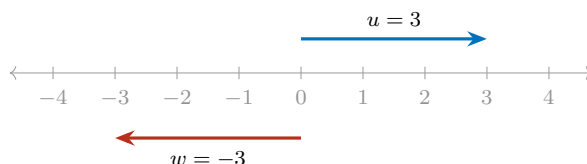
The pair to dwell on is **speed** and **velocity**. Speed is a scalar: 60 mph. Velocity is a vector: 60 mph *due north*. Speed is exactly the magnitude of the velocity vector — that is, its norm — which is why the two words are not synonyms and why a car going round a bend at constant speed is nonetheless accelerating.

5. One-dimensional vectors

Nothing above required two dimensions. In one dimension — on the number line — a vector is a signed number, and the two attributes are still both present:

- the **sign** carries the direction: positive points one way along the line, negative the other;
- the **absolute value** carries the magnitude.

So $u = 3$ and $w = -3$ have the same magnitude and opposite directions. And the one-dimensional norm is $\|u\| = \sqrt{u^2} = |u|$ — absolute value was your first norm all along.



6. Notation, and where vectors sit set-theoretically

There are several standard ways to denote a vector symbolically. Using the letter v purely as an example — *any* letter may be used:

Notation	Reading
$\mathbf{v}$	boldface — common in applied and computational writing
$\vec{v}$	arrow overhead — common in physics and calculus texts
v	plain — common in pure linear algebra, where context suffices

In this class we will use a mixture of all three, and context will tell you which object you are looking at. Getting comfortable reading all three is part of the course: you will meet all of them in the reference texts.

The set-theoretic statement

Until now we have said “the plane” in words. Here is its name. To say that $\mathbf{v}$ is a two-dimensional vector, we write

$$\mathbf{v} \in \mathbb{R}^2, \quad \text{where} \quad \mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}.$$

Read aloud: “ $\mathbf{v}$ is in $\mathbb{R}$ -two.” The superscript 2 is the dimension of §1 — two coordinates, each free to be any real number, chosen independently. From here on, $\mathbb{R}^2$ replaces “the plane” as our standard name for the space a two-dimensional vector lives in.

The numerical convention for this class

We write a vector in brackets, listing its coordinates in order:

$$\mathbf{v} = [x, y], \quad \text{for instance} \quad \mathbf{v} = [1, 2].$$

Order matters: $[1, 2] \neq [2, 1]$.

Looking ahead. At this stage of the course, a vector *is* a point of the plane together with the convention that its tail sits at the origin: the head is (x, y) and the arrow is drawn from $(0, 0)$. This is a temporary simplification and we will discard it once you are fluent — a vector is really a *displacement*, so the same vector may be drawn with its tail anywhere, which is precisely what makes the head-to-tail picture in §7 legitimate.

7. Addition and subtraction

Many operations on vectors await us. Here are the first two.

Definition 3 (Addition and subtraction in $\mathbb{R}^2$). Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{u} = [x_1, y_1]$ and $\mathbf{v} = [x_2, y_2]$. Then

$$\mathbf{u} + \mathbf{v} = [x_1 + x_2, y_1 + y_2], \quad \mathbf{u} - \mathbf{v} = [x_1 - x_2, y_1 - y_2].$$

Both are **componentwise**: the first coordinates interact only with each other, and likewise the second. Nothing mixes across slots.

Now notice the *shape* of these two operations, exactly as we did for the norm in §2. Each one is a **binary operator**: it consumes *two* vectors and returns a vector.

$$+ : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad - : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2.$$

Read the left-hand side as “a pair of vectors from $\mathbb{R}^2$ goes in.” Contrast this with the norm, which was *unary* — one vector in, one *number* out. Addition and subtraction are binary, and they never leave $\mathbb{R}^2$: two vectors in, one vector out.

Example 3 (Both operations on one pair). Let $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$. Then

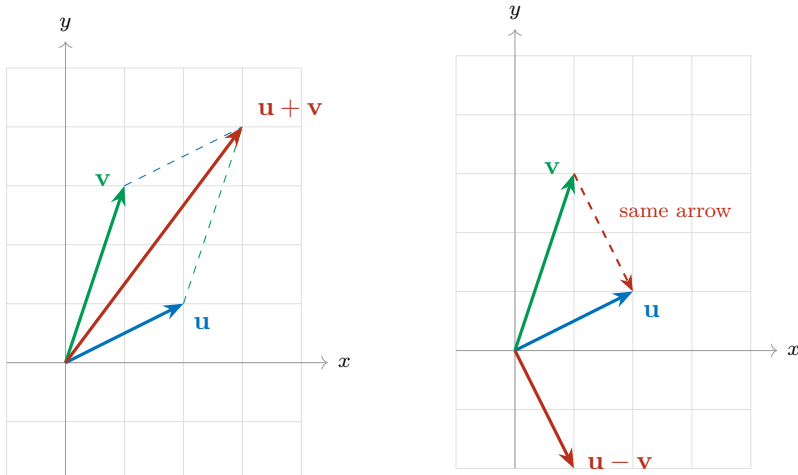
$$\mathbf{u} + \mathbf{v} = [2 + 1, 1 + 3] = [3, 4], \quad \mathbf{u} - \mathbf{v} = [2 - 1, 1 - 3] = [1, -2].$$

And since we know how to take a norm: $\|\mathbf{u} + \mathbf{v}\| = \sqrt{9 + 16} = 5$ — the clean vector from §2, arrived at by addition.

What the pictures say

Addition is the *parallelogram rule*: place $\mathbf{u}$ and $\mathbf{v}$ tail-to-tail at the origin, complete the parallelogram, and $\mathbf{u} + \mathbf{v}$ is the diagonal from the origin. Equivalently — *head to tail* — walk along $\mathbf{u}$, then from where you land walk along $\mathbf{v}$; the sum is the single arrow from start to finish.

Subtraction is addition of the opposite, $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$, and it has its own reading: $\mathbf{u} - \mathbf{v}$ is the arrow *from the head of $\mathbf{v}$ to the head of $\mathbf{u}$* . It is the displacement that takes you from $\mathbf{v}$ to $\mathbf{u}$, which is why differences of vectors are how we will measure how far apart two vectors are.



In the right-hand figure the dashed arrow and the solid arrow labelled $\mathbf{u} - \mathbf{v}$ are the *same vector* drawn in two places: same magnitude, same direction, different tail. That is the translation-invariance flagged in §6, seen for the first time.

8. Scalar multiplication

Addition and subtraction combine a vector with another vector. The next operation combines a vector with a *scalar* — one of the plain numbers of §4, carrying magnitude but no direction.

Definition 4 (Scalar multiplication in $\mathbb{R}^2$). Let $\mathbf{v} = [x, y] \in \mathbb{R}^2$ and let $\alpha \in \mathbb{R}$. Then

$$\alpha \mathbf{v} = [\alpha x, \alpha y].$$

Componentwise again: α multiplies *every* coordinate.

This operation is binary as well, but its two inputs are of *different kinds* — a number and a vector, not two vectors:

$$\mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2.$$

A scalar and a vector go in; a vector comes out. That mixed signature is why α is called a *scalar* in the first place: its job is to set the scale of the vector.

Example 4 (Scaling the running vector). Let $\mathbf{v} = [1, 2]$, our vector from §1, with $\|\mathbf{v}\| = \sqrt{5}$.

$$2\mathbf{v} = [2, 4], \quad \frac{1}{2}\mathbf{v} = [\frac{1}{2}, 1], \quad -\mathbf{v} = (-1)\mathbf{v} = [-1, -2].$$

Their norms are

$$\|2\mathbf{v}\| = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}, \quad \|\frac{1}{2}\mathbf{v}\| = \sqrt{\frac{1}{4} + 1} = \frac{\sqrt{5}}{2}, \quad \|-\mathbf{v}\| = \sqrt{1 + 4} = \sqrt{5}.$$

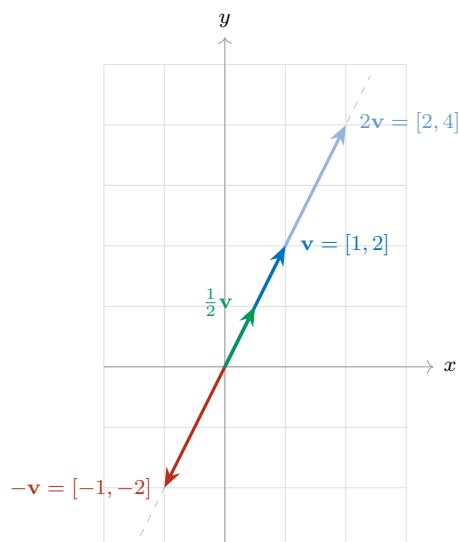
So $2\mathbf{v}$ is twice as long as $\mathbf{v}$, $\frac{1}{2}\mathbf{v}$ is half as long, and $-\mathbf{v}$ is exactly as long as $\mathbf{v}$ but points the opposite way.

Remark. Read off the pattern: scaling by α multiplies the magnitude by $|\alpha|$,

$$\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|,$$

and the *sign* of α decides the direction — positive keeps it, negative reverses it. That is the same sign-carries-direction rule you saw on the number line in §5, now acting on a vector in the plane. Taking $\alpha = 0$ collapses the vector to $[0, 0]$, the **zero vector**, which has no direction at all.

Geometrically, scalar multiplication never turns a vector off its own line: every multiple of $\mathbf{v}$ lies on the single straight line through the origin and the head of $\mathbf{v}$. Scaling slides the arrowhead along that line — stretching it out, pulling it in, or flipping it through the origin — and nothing else.



9. Practice set

For each pair below: **(a)** draw $\mathbf{u}$ and $\mathbf{v}$ on one set of axes, tails at the origin, arrowheads marked; **(b)** compute the indicated sum or difference componentwise; **(c)** draw the resulting vector on the same axes.

P1. $\mathbf{u} = [3, 1]$, $\mathbf{v} = [1, 2]$. Compute $\mathbf{u} + \mathbf{v}$.

P2. $\mathbf{u} = [4, 1]$, $\mathbf{v} = [1, 3]$. Compute $\mathbf{u} - \mathbf{v}$.

P3. $\mathbf{u} = [-2, 3]$, $\mathbf{v} = [3, 1]$. Compute $\mathbf{u} + \mathbf{v}$.

P4. $\mathbf{u} = [2, -2]$, $\mathbf{v} = [-3, -1]$. Compute $\mathbf{u} - \mathbf{v}$.

Optional: compute the norm of each resultant.

Watch the signs in P4: both inputs have a negative component, so $2 - (-3) = 5$ and $-2 - (-1) = -1$.

Answer key

#	$\mathbf{u}$	$\mathbf{v}$	Result	$\ \text{result}\ $
P1	$[3, 1]$	$[1, 2]$	$\mathbf{u} + \mathbf{v} = [4, 3]$	5
P2	$[4, 1]$	$[1, 3]$	$\mathbf{u} - \mathbf{v} = [3, -2]$	$\sqrt{13}$
P3	$[-2, 3]$	$[3, 1]$	$\mathbf{u} + \mathbf{v} = [1, 4]$	$\sqrt{17}$
P4	$[2, -2]$	$[-3, -1]$	$\mathbf{u} - \mathbf{v} = [5, -1]$	$\sqrt{26}$